SKD 51



V_{RSM} V	V_{RRM}, V_{DRM} V	I_D = 50[1] A (full conduction) (T_c = 127 °C)
500	400	SKD 51/04
900	800	SKD 51/08
1300	1200	SKD 51/12
1500	1400	SKD 51/14
1700	1600	SKD 51/16
1900	1800	SKD 51/18

Power Bridge Rectifiers

SKD 51

Features

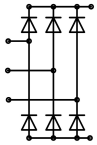
- Glass passivated silicon chips
- Fast-on terminals for pcb solder or plug on connections
- Sturdy insulated metal base plate
- Low thermal impedance through use of direct copper bonded aluminum substrate
- Blocking voltage up to 1800V
- High surge currents
- UL recognized, file no. E63 532

Typical Applications*

- Three phase rectifier for power supplies
- Input rectifier for variable frequency drives
- Rectifier for DC motor field supplies
- Battery charger rectifiers
- Recommended snubber network: RC: 0.1 µF, 50 Ω (P_R = 1 W)

Symbol	Conditions	Values	Units
I_D	T_c = 127 °C	50	A
	T_a = 45 °C; isolated [2]	7	A
	T_a = 45 °C; chassis [3]	18	A
	T_a = 45 °C; R4A/120	27	A
	T_a = 45 °C; P5A/100	31	A
I_{FSM}	T_{vj} = 25 °C; 10 ms	775	A
	T_{vj} = 150 °C; 10 ms	700	A
i^2t	T_{vj} = 25 °C; 8,3 ... 10 ms	3000	A²s
	T_{vj} = 150 °C; 8,3 ... 10 ms	2450	A²s
V_F	T_{vj} = 25 °C; I_F = 75 A	max. 1,45	V
$V_{(TO)}$	T_{vj} = 150 °C	max. 0,8	V
r_T	T_{vj} = 150 °C	max. 8,5	mΩ
I_{RD}	T_{vj} = 25 °C; V_{DD} = V_{DRM}; V_{RD} = V_{RRM}	max. 0,2	mA
	T_{vj} = 150 °C; V_{RD} = V_{RRM}	4	mA
t_{rr}	T_{vj} = 25 °C; I_F = I_R = 1A	5	µs
$R_{th(j-c)}$	per diode	1,1	K/W
	total	0,183	K/W
$R_{th(c-s)}$	total	0,1	K/W
$R_{th(j-a)}$	isolated [2] (chassis [3])	9 (3,15)	K/W
T_{vj}		- 40 ... +150	°C
T_{stg}		- 40 ... +125	°C
V_{isol}	a. c. 50 Hz; r.m.s.; 1 s / 1 min.	3600 (3000)	V
M_s	to heatsink	4,5 ± 15 %	Nm
M_t			
m		97	g
Case		G 51	

1) For solder connection. Permissible current for plug connection see DIN IEC 760E and DIN 46249 part 1
2) Freely suspended or mounted on an insulator
3) Mounted on a painted metal sheet of min. 250 x 250 x 1 mm



SKD

© by SEMIKRON

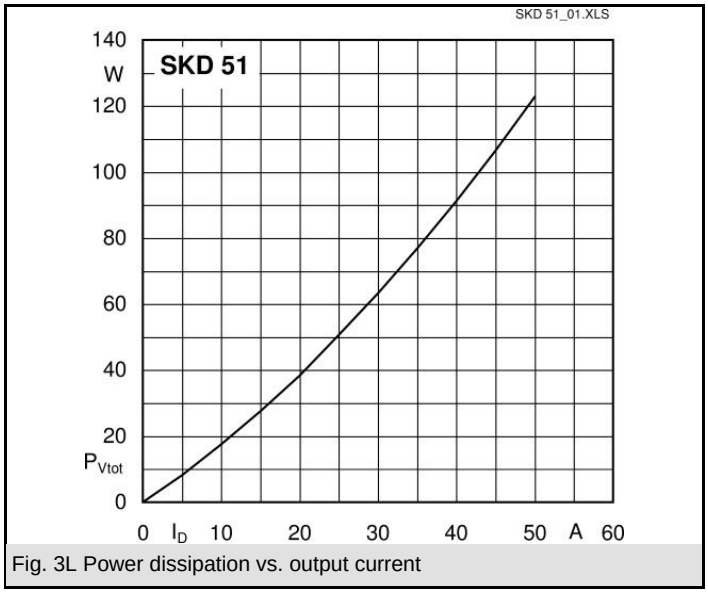


Fig. 3L Power dissipation vs. output current

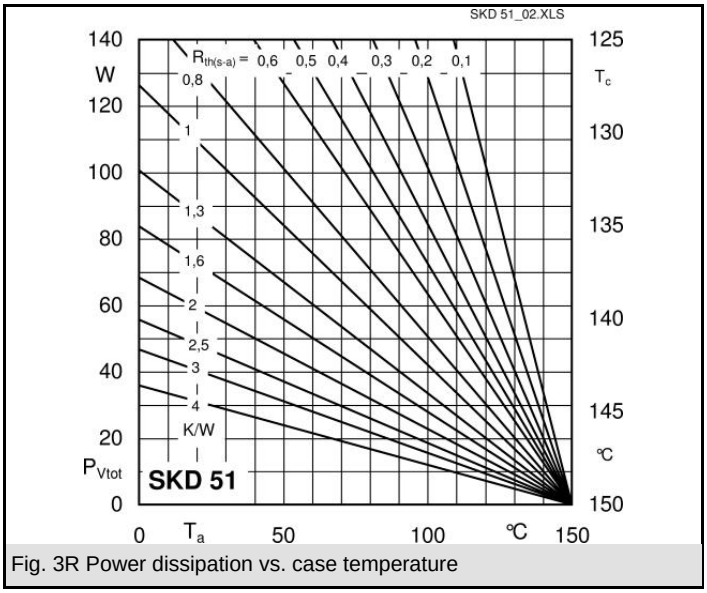


Fig. 3R Power dissipation vs. case temperature

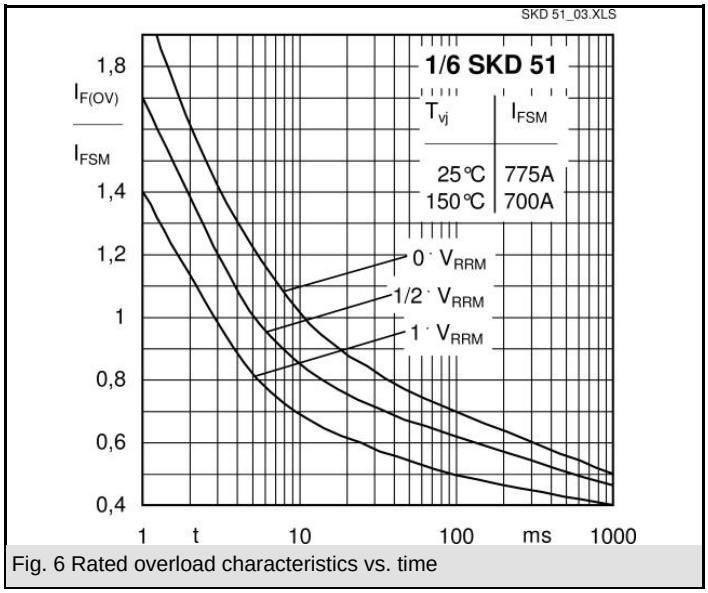
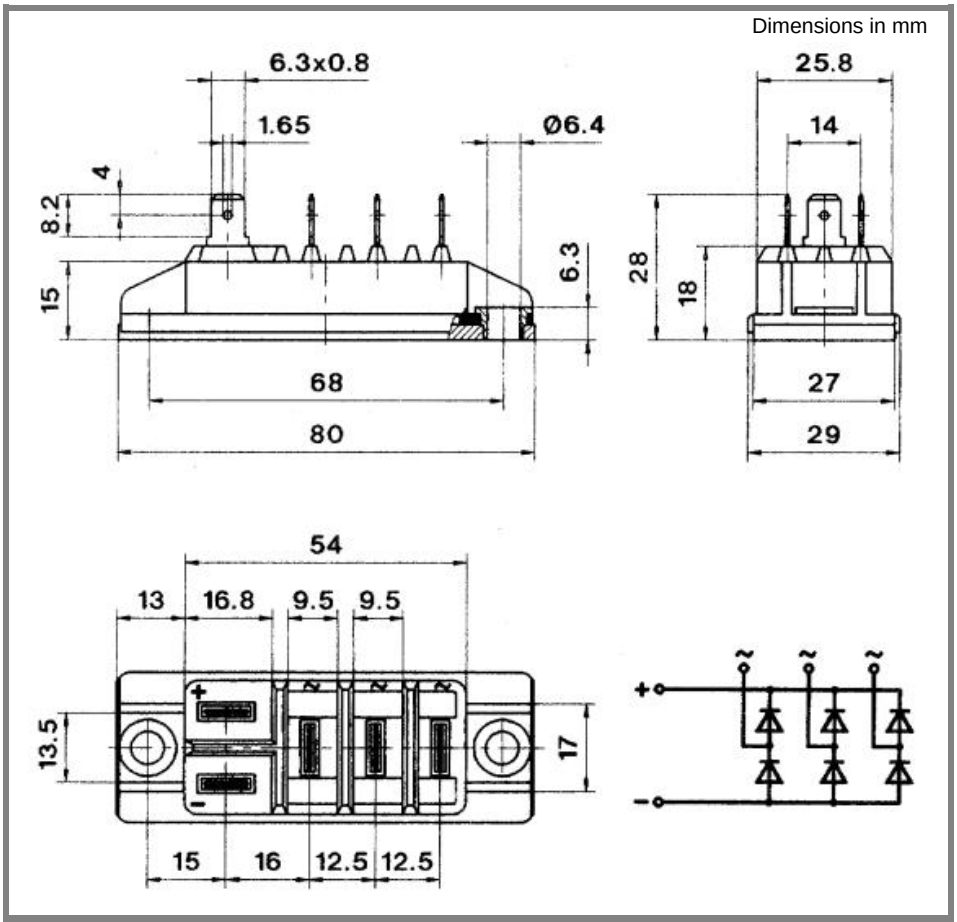


Fig. 6 Rated overload characteristics vs. time

 © by SEMIKRON

SKD 51



Case G 51

* The specifications of our components may not be considered as an assurance of component characteristics. Components have to be tested for the respective application. Adjustments may be necessary. The use of SEMIKRON products in life support appliances and systems is subject to prior specification and written approval by SEMIKRON. We therefore strongly recommend prior consultation of our personal.

 © by SEMIKRON